

Equations with Brackets and Fractions

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|-------|--------------------------------------|
| 1. 3 | 7. 6 |
| 2. 9 | 8. 7 |
| 3. 16 | 9. 6 |
| 4. 12 | 10. 15 |
| 5. 4 | 11. 30 |
| 6. 4 | 12. (a) the correct solution = 10, — |
-

What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 13 tagged checks.

1. Solve $3(x + 4) = 21$.

Solution: Distribute across both terms: $3x + 12 = 21$. Subtract twelve from both sides to get $3x = 9$, then divide by three. $x = 3$

2. Solve $5(x - 2) = 35$.

Solution: Distributing gives $5x - 10 = 35$. Add ten to both sides: $5x = 45$. Divide by five. $x = 9$

3. Solve $\frac{x}{4} + 3 = 7$.

Solution: Subtract three from both sides first, leaving one quarter of the unknown equal to four; then multiply both sides by four. $x = 16$

4. Solve $\frac{2x}{3} = 8$.

Solution: Multiply both sides by three to clear the denominator: $2x = 24$. Divide by two. $x = 12$

5. Solve $\frac{x+6}{2} = 5$.

Solution: The fraction bar groups the whole numerator, so multiplying both sides by two gives $x+6 = 10$ — the right side is doubled too, which is the step most often missed. Subtract six.

$$x = 4$$

6. Solve $2(3x-1) + 4 = 26$.

Solution: Distribute first: $6x - 2 + 4 = 26$, so $6x + 2 = 26$. Subtract two from both sides: $6x = 24$. Divide by six.

$$x = 4$$

7. Solve $\frac{x}{3} + \frac{x}{6} = 3$.

Solution: The least common denominator is six, so multiply every term by six: $2x + x = 18$. Collect on the left to get $3x = 18$, then divide by three.

$$x = 6$$

8. Solve $\frac{x+1}{2} + \frac{x-3}{4} = 5$.

Solution: Multiply every term by four, treating each numerator as a bracket.

$$2(x+1) + (x-3) = 20$$

$$2x + 2 + x - 3 = 20$$

$$3x - 1 = 20$$

$$3x = 21$$

Divide both sides by three.

$$x = 7$$

9. Solve $4(2x+3) = 3(x+14)$.

Solution: Expand both brackets before collecting anything: $8x + 12 = 3x + 42$. Subtract $3x$ from both sides: $5x + 12 = 42$. Subtract twelve: $5x = 30$. Divide by five.

$$x = 6$$

10. Solve $\frac{3x}{5} - 2 = \frac{x}{5} + 4$.

Solution: Multiply every term by five: $3x - 10 = x + 20$. Subtract x from both sides: $2x - 10 = 20$. Add ten: $2x = 30$. Halve.

$$x = 15$$

11. Solve $\frac{2(x-6)}{3} = \frac{x+2}{2}$.

Solution: The least common denominator is six, so multiply both sides by it, keeping each numerator grouped.

$$4(x-6) = 3(x+2)$$

$$4x - 24 = 3x + 6$$

$$x - 24 = 6$$

Add twenty-four to both sides.

$$x = 30$$

12. **Challenge.** Nia's work on $3(x-4) = 18$. (a) Solve correctly. (b) What went wrong?

Solution (a): The outside factor multiplies both terms inside: $3x - 12 = 18$. Add twelve to both sides: $3x = 30$. Divide by three.

$$x = 10$$

Solution (b) — instructor-judged. Nia distributed to the variable term only and left the constant inside untouched, so her second line is wrong and everything after it inherits the error. The distributive property applies to *every* term inside the bracket. Substituting her value into the original equation does not give the right-hand side, while the correct value gives eighteen exactly. Full credit requires naming the every-term rule.